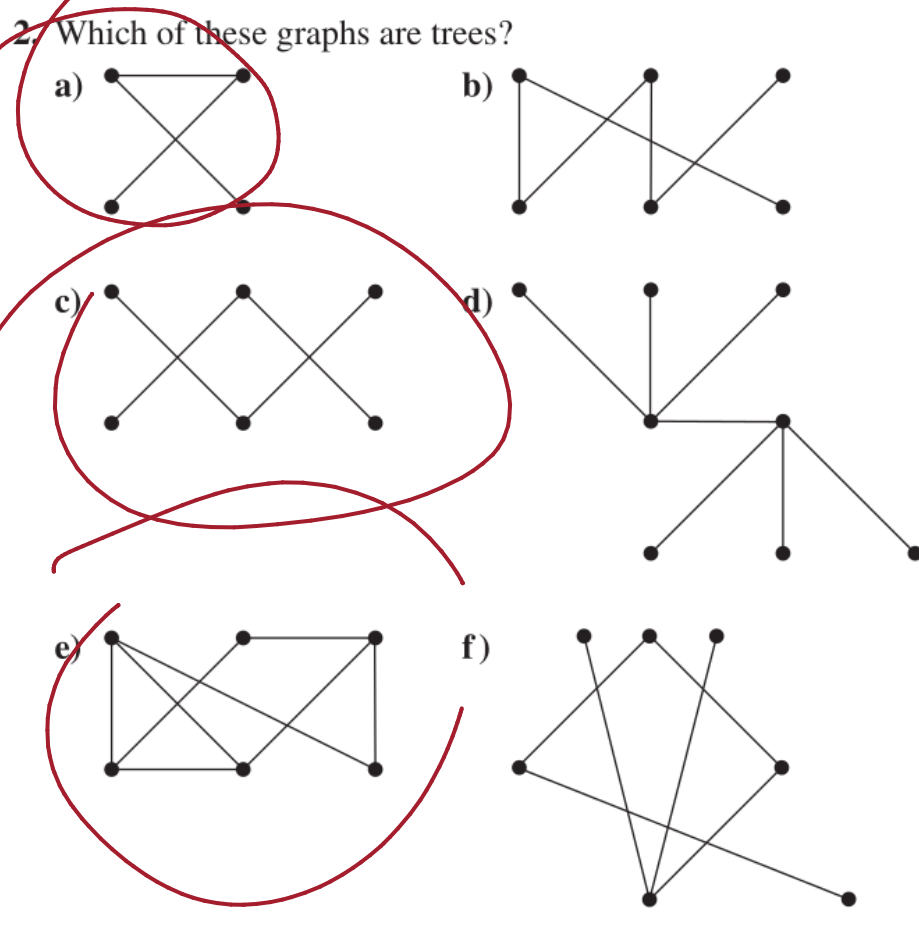
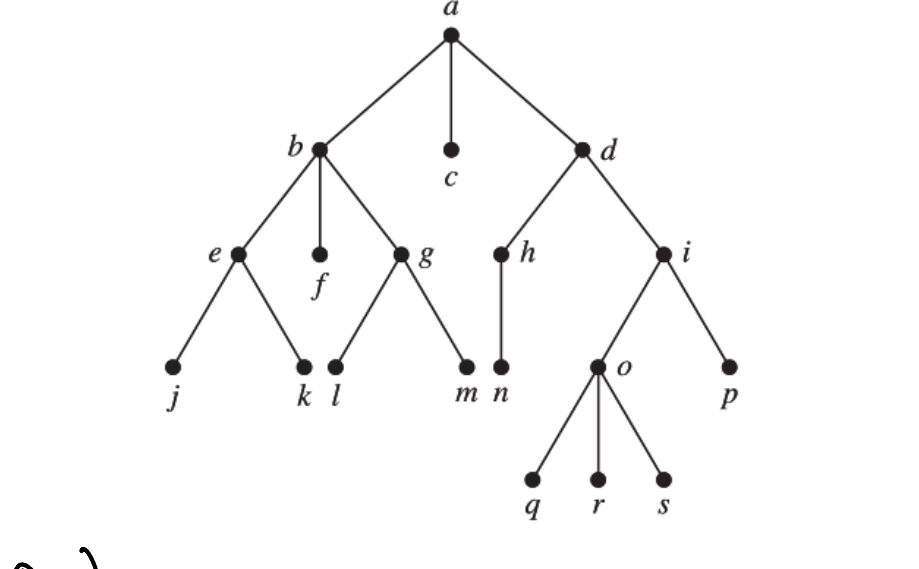




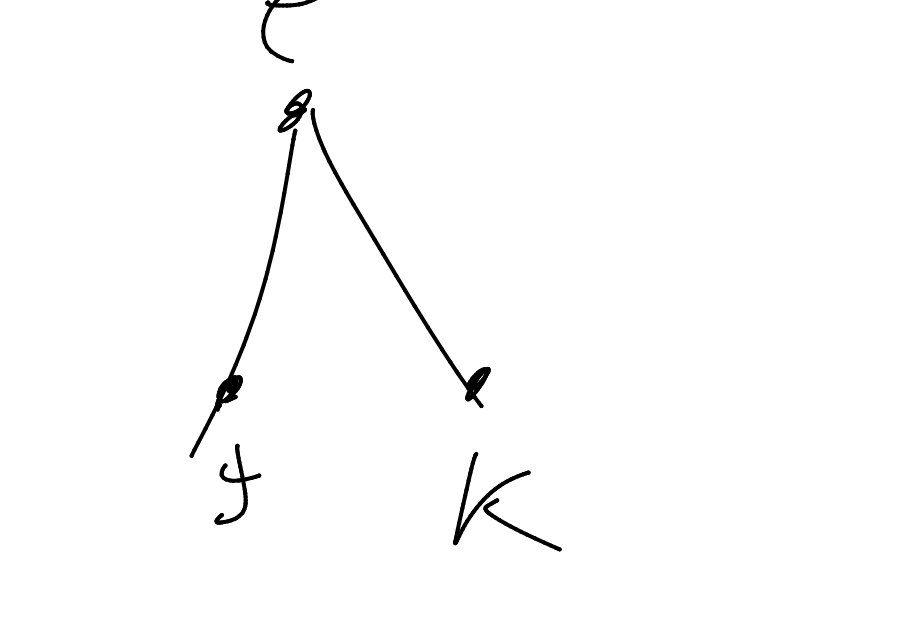
a) tree
 c) not a tree
 f) not a tree

- Which vertex is the root?
 - Which vertices are internal?
 - Which vertices are leaves?
 - Which vertices are children of j ?
 - Which vertex is the parent of k ?
 - Which vertices are siblings of e ?
 - Which vertices are ancestors of m ?
 - Which vertices are descendants of h ?
4. Answer the same questions as listed in Exercise 3 for the rooted tree illustrated.

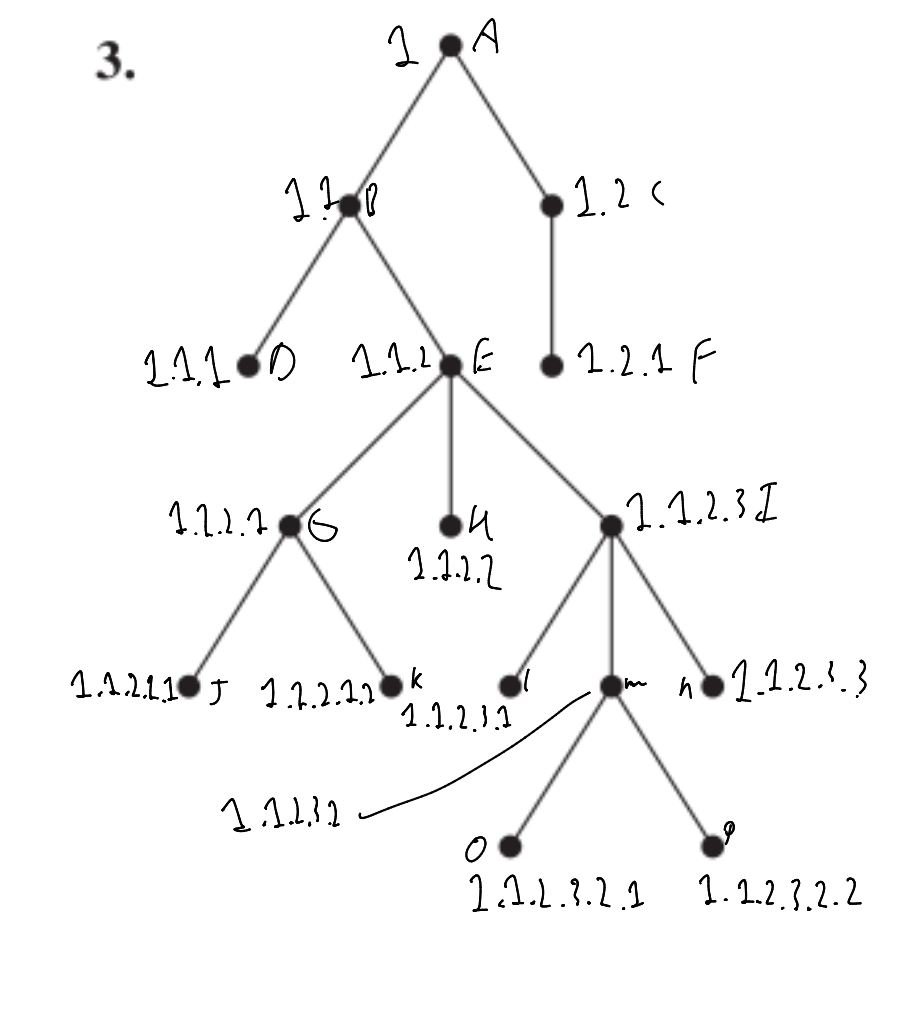


a) a
 b) b, d, e, g, h, i, o
 c) j, k, l, m, n, p, q, r, s
 d) none
 e) j
 f) p
 g) g, b, a
 h) e, f, g, j, k, l, m

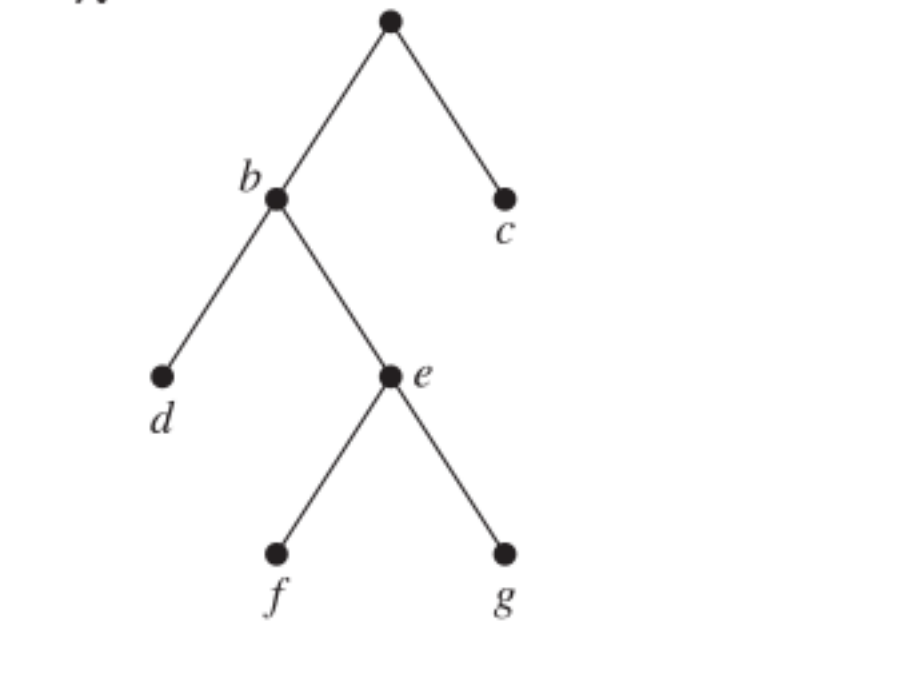
10. Draw the subtree of the tree in Exercise 4 that is rooted at
- a.
 - c.
 - e.



In Exercises 1–3 construct the universal address system for the given ordered rooted tree. Then use this to order its vertices using the lexicographic order of their labels.



In Exercises 7–9 determine the order in which a preorder traversal visits the vertices of the given ordered rooted tree.



a, b, d, e, f, g, c

10. In which order are the vertices of the ordered rooted tree in Exercise 7 visited using an inorder traversal?

d, b, f, e, g, a, c

13. In which order are the vertices of the ordered rooted tree in Exercise 7 visited using a postorder traversal?

d, f, g, e, b, c, a